A search for waste heat from Dyson spheres around nearby stars in archival infrared surveys

Charles C. Norton

Abstract

A structure that intercepts a star's light re-radiates it as heat, so a Dyson sphere shows as infrared emission beyond what the star alone gives. We test nearby stars for that emission in ten ways with archival surveys. None of the 279 optically dark Gaia DR3 astrometric companions of $0.5\,M_\odot$ or more within 200 pc re-radiates more than 6 per cent of a main-sequence star's light at 300 K (median bound 0.3 per cent), nor do the 191 companions of $0.3\,M_\odot$ or more at 3 to 30 au implied by Hipparcos-Gaia proper-motion anomalies within 50 pc (median bound 0.39 per cent). Stacked against matched field stars, the WISE colours of habitable-zone planet hosts within 100 pc limit the mean fraction of their light intercepted by 300 K swarms to 1.5×10^{-5}. No bright, cold, moving AllWISE source without a Gaia star is an enclosed star, no star within 50 pc has a 30 to 100 K shell in the AKARI far-infrared survey, none of 203 stars within 15 pc brightened at 4.6 μm over the decade of NEOWISE by more than about 9 per cent, and no star that vanished from the Palomar plates reappears with the 12 μm emission an enclosure requires. SPHEREx spectra trace the excess of earlier Dyson-sphere candidates to background galaxies and a radio source, and put the excesses of 37 extreme mid-infrared stars at 486 to 2133 K, the temperatures of warm dust and cool companions, none below 400 K within the 5 μm reach of the spectra. Each test's reach is stated: the enclosed-star search finds a 300 K shell around a star of $0.01\,L_\odot$ to 210 pc, the AKARI match finds a 30 to 100 K shell holding 0.2 per cent of a solar luminosity at 50 pc, and a vanished source of $R = 19$ enclosed at 200 to 600 K would have been detected by WISE.

Keywords: Technosignatures, Dyson spheres, Infrared: stars, Circumstellar matter, Solar neighbourhood

1. Introduction

Dyson proposed that a civilization using much of its star's output would surround the star with collectors and re-radiate the energy as waste heat, and that the heat, at a few hundred kelvin, would make the system a source of infrared emission that its starlight does not explain [1]. Searches have since used each infrared survey in turn: IRAS for whole-sky limits on spheres at 100 to 600 K [2], WISE for galaxies whose stellar light is converted to mid-infrared emission [3, 4], and Gaia with 2MASS and WISE for partial spheres around individual Milky Way stars [5, 6, 11]. A detected excess has natural rivals: debris discs, red companions, background galaxies and young stars embedded in their clouds.

The nearest stars allow tests that need a known distance, mass or history. A companion whose mass Gaia's astrometry measures but whose light it does not see is either a dim remnant or a star enclosed; a structure built within a decade brightens its star in the infrared from one epoch to the next; a star enclosed during the twentieth century would vanish from the Palomar plates and

reappear in the mid-infrared. This paper applies these tests and the stacking and spectroscopic ones to archival surveys, and states each result as a fraction of a star's light re-radiated at the temperature tested.

1.1. Contributions

1. Limits on the fraction of light re-radiated at 300 K by 279 Gaia DR3 dark companions and by 191 companions inferred from proper-motion anomalies.
2. A stacked limit on warm swarms around habitable-zone planet hosts, and searches for fully enclosed moving stars, cold far-infrared shells, decade-long brightening and stars that vanished from the optical sky into the infrared.
3. SPHEREx 0.75 to 5 μm spectra of published Dyson-sphere candidates against matched controls, which place every excess in a natural class.

2. Tests

Table 1 lists the ten tests, their samples and results. Each compares a measured infrared flux with what the star or its catalogued companions would give, and expresses the difference as the fraction of the star's light that a black body at the tested temperature would need to re-radiate.

Dark astrometric companions. Gaia DR3's orbital solutions [12] give the mass of each companion and the flux ratio its orbit allows in the G band. Within 200 pc, 279 companions of $0.5\,M_\odot$ or more are dark, contributing less than a third of the light of a main-sequence star of their mass. Inside a complete sphere such a star would re-radiate its main-sequence luminosity as a 300 K black body, whose 12 and 22 μm flux at the Gaia distance we compare with the AllWISE [13] counterpart within $3''$.

Companions from proper-motion anomalies. The difference between a star's Hipparcos-Gaia long-term proper motion and its Gaia proper motion gives the least mass of a companion anywhere between 3 and 30 au [14]. For the 191 Hipparcos stars within 50 pc whose anomaly, at a signal-to-noise ratio of 5, requires a companion of $0.3\,M_\odot$ or more, a star of that mass inside a shell at 200, 300 or 600 K would add predictable 12 and 22 μm flux; the star's $K_s - W3$ and $K_s - W4$ colours, against the 15 sample stars nearest in $G - K_s$, bound the fraction re-radiated. An excess is read again from the WISE All-Sky release, which processed the same cryogenic 12 and 22 μm exposures apart from AllWISE, against the same neighbours.

Stacking at habitable-zone hosts. A swarm intercepting a fraction f of the starlight at 300 K adds 12 and 22 μm flux in proportion to f, so a population of modest swarms shows as a mean colour excess. Hosts of planets receiving 0.32 to 1.78 times the Earth's insolation [15] within 100 pc in the NASA Exoplanet Archive [16], and the other hosts as comparison, each receive up to 25 field stars matched in Gaia colour, magnitude and parallax. The inverse-variance mean of the hosts' $W1 - W3$ and $W1 - W4$ excesses over their controls, against the same statistic with a control standing for its host, limits the mean f.

Fully enclosed stars. A shell that absorbs all of a star's light re-radiates it at the shell's temperature: at 300 K around an M dwarf of $0.01\,L_\odot$ at 100 pc it gives $W3 \approx 6.4$ with $W2 - W3 \approx 5.7$, and Gaia sees nothing. Of the 36603 AllWISE point sources with $W3 < 8$ at a signal-to-noise ratio above 10 and $W2 - W3 > 1.5$, we keep those that CatWISE2020 [17] finds moving faster than $40\,\mathrm{mas\,yr^{-1}}$ at 5σ, as a star within 100 pc would, with no Gaia DR3 source within $3''$, clean flags and unsaturated $W1$ and $W2$, and check each against SIMBAD and for young stars, cores

and clouds within 30′. The $W3 < 8$ cut reaches a 300 K shell around a star of 0.001, 0.01 and 0.1 $L_\odot$ to 67, 210 and 670 pc, and shells at 200 and 500 K to within 20 per cent of those distances; the motion cut keeps stars whose tangential velocity exceeds 19 km s^{-1} at 100 pc and 40 km s^{-1} at 210 pc, about three quarters of the disc population at the first and a third at the second.

Cold shells. A structure at 30 to 100 K peaks beyond the WISE bands, and a main-sequence photosphere at 65 to 160 μm lies far below the AKARI Far-Infrared Surveyor's bright source catalogue [18]. We match the 27394 Gaia DR3 main-sequence stars within 50 pc to that catalogue within 20″, and take the chance count from the same match with every position moved by 3′. The catalogue's 0.55 Jy limit at 90 μm corresponds, for a shell at 30 to 100 K, to 0.2 to 0.6 per cent of a solar luminosity at 50 pc and 0.03 to 0.09 per cent at 20 pc; for the M dwarfs that make up most of the sample, of 0.01 $L_\odot$ at 10 pc, it needs 1 to 2 per cent of the star's light, so the test bounds complete or nearly complete cold shells around the M dwarfs and partial ones around the solar-type stars.

Decade trends. A structure assembled during 2014 to 2024 at 600 to 1000 K brightens its star year by year, more at 4.6 than at 3.4 μm. For the 298 Gaia DR3 stars within 15 pc with $W1$ of 8 to 12.5, the NEOWISE-R single exposures [19] from December 2013 to August 2024 give $W1$ and $W2$ light curves differential against field stars on the same frames, and a linear trend per decade is judged against stars of similar brightness, since the photometry drifts with brightness.

Vanished in the optical, present in the infrared. A star enclosed after the 1950s leaves the first Palomar survey's plates and returns at its enclosure's temperature: at 300 K a star of $R = 19$ appears near $W2 = 16.8$ and $W3 = 11.1$, at 200 K near $W3 = 11.6$ and $W4 = 9.0$, and at 600 K near $W2 = 14.1$, against AllWISE's 5σ limits of about 17.5, 11.6 and 8.0 in $W2$, $W3$ and $W4$, so an enclosure at any of those temperatures of a source brighter than $R \approx 19$, which is most of the catalogue, would be detected in at least one band. The archive of vanishing objects [20] built for the VASCO project [21] lists Palomar sources absent from Pan-STARRS and Gaia; we keep those whose only counterpart is infrared and red in $W1 - W2$, away from the ecliptic, with a fixed CatWISE2020 counterpart and no Legacy Surveys source [22] to $r = 24$.

Breakthrough Listen targets. The FGK stars of Breakthrough Listen's nearby-star sample [23] observed in a set of 2016 L-band sessions, 170 in all, are tested for $K_s - W3$ or $K_s - W4$ above 0.25 mag at 5σ with clean photometry.

SPHEREx spectra. SPHEREx images the sky in 102 bands from 0.75 to 5 μm [24]. We measure each target, carried by proper motion to the epoch of the images, by a point-spread fit after the zodiacal model in every image covering it, and the same for two Gaia stars of matching colour, magnitude and distance that WISE does not detect at 12 μm. Normalised at 1.0 to 1.6 μm, a target's excess over its controls at 2 to 3, 3 to 4 and 4 to 5 μm is judged against one control measured against the other. The targets are candidates of Project Hephaistos [6] and the 53 FGK stars with the most extreme Gaia-predicted $W2$ excesses of Contardo and Hogg [11], of which 51 have controls. A black body fitted from 1.6 to 5 μm gives the temperature of each excess and the fraction of the star's light it holds. The Hephaistos targets are ten candidates of that search and one M dwarf with a warm excess, eleven in all.

3. Results

Companions. No dark astrometric companion shows the 12 or 22 μm flux that a star of its mass inside a 300 K sphere would give: the measured flux bounds the re-radiated fraction below 6 per cent for every one of the 279 and below 0.3 per cent for half of them. The proper-motion

Table 1: The ten tests. f is the fraction of a star's light intercepted and re-radiated at the temperature tested.

Test	Sample	Result
Dark companions	279 Gaia DR3 orbital companions $\geq 0.5\,M_\odot$, < 200 pc	none above 6 per cent of its light at 300 K; median bound 0.3 per cent
Proper-motion companions	191 stars < 50 pc, companion $\geq 0.3\,M_\odot$ at 3–30 au	median bound 0.39 per cent at 300 K; four 3σ excesses, all explained
Habitable-zone stacking	69 hosts ($W3$), 63 ($W4$), < 100 pc, 25 controls each	mean $f < 1.5 \times 10^{-5}$ (12 μm) and 8.6×10^{-6} (22 μm) at 300 K
Fully enclosed stars	36603 cold AllWISE sources	72 clean movers without Gaia stars, all among young stars, cores and clouds
Cold shells	27394 stars < 50 pc, AKARI FIS	35 matches against 26 by chance; clouds and known young discs
Decade trends	203 stars < 15 pc, NEOWISE 2013–2024	none brightens at 4.6 μm at 5σ; bound about 9 per cent
Vanished stars	5893 red infrared-only counterparts of vanishing Palomar sources	four red CatWISE sources left, none detected at 12 μm
Breakthrough Listen targets	170 FGK stars	three excesses: a debris disc, a blend, a catalogued 22 μm excess
SPHEREx, Hephaistos	11 candidates, 2 controls each	background galaxies and a radio source
SPHEREx, Contardo and Hogg	51 stars, 2 controls each	37 excesses at 486–2133 K, none below 400 K

companions give a median bound of 0.39 per cent at 300 K, and 154 of 189 with both bands are bounded below 1 per cent. Three of their four 3σ colour excesses are a diffraction spike from a neighbour, the photosphere of an M0 star and the white-dwarf companion of 56 Per. The fourth, 0.14 mag at 12 μm (3.3σ) in the F6 V star HD 59099, whose astrometry (RUWE 11.9) marks the companion, belongs to the AllWISE measurement: the All-Sky release puts the star 0.11 mag fainter at 12 μm and 0.07 mag fainter at 22 μm, where the two releases agree for the sample's stars to a median of 0.002 mag with a scatter of 0.007 mag, and against the same neighbours its All-Sky excess is +0.02 mag at 12 μm and nothing at 22 μm.

Habitable-zone hosts. The 69 habitable-zone hosts with clean 12 μm photometry and controls show a mean $W1 - W3$ excess of $+0.0004 \pm 0.0045$ mag over their controls, the 124 other hosts -0.0011 ± 0.0036 mag, and controls standing for their hosts $+0.0018 \pm 0.0006$ mag; at 22 μm the 63 habitable-zone hosts give -0.0005 ± 0.0094 mag, the 88 other hosts -0.0060 ± 0.0106 mag, and controls standing for their hosts $+0.0123 \pm 0.0017$ mag. The control null is offset from zero in both bands, by 0.0018 mag at 12 μm and 0.012 mag at 22 μm: a control measured against the mean of the others is systematically redder than that mean, which the hosts share, so the hosts' excesses are read against the null and fall below it by 0.3σ and 1.3σ. A 300 K swarm therefore intercepts on average less than 1.5×10^{-5} of a habitable-zone host's light (8.6×10^{-6} at 22 μm). The one host above 3σ, TOI-2257, has a galaxy 7.3″ away.

Enclosed stars and cold shells. Of the 36603 cold AllWISE sources, 7817 move at 5σ, 1491 of these have no Gaia source, and 72 have clean flags below saturation; every one of the 72 lies among young stars, dense cores or clouds, where variability and nebulosity bias CatWISE motions. Within 50 pc, 35 stars have an AKARI 90 μm source within 20″ against 26 when the positions are moved;

the matches within 8″ are alignments with clouds in fields of 43 to 349 sources per square degree (V854 Ori, V1497 Cyg, HD 324218) or known young discs (HD 98800 in the TW Hya association, and HD 164249 A in the β Pictoris group at 7×10^{-4} of its light).

Decade trends and vanished stars. Of the 203 stars within 15 pc with both NEOWISE light curves, none brightens at 4.6 μm at 5σ, the most extreme trend being -2.8σ, and the median slope error of 0.018 mag per decade bounds any gain at about 9 per cent of the 4.6 μm flux. Of the 5893 red infrared-only counterparts of vanished Palomar sources, those near the ecliptic are asteroids caught in single NEOWISE exposures and those detected in one exposure without a CatWISE source are transients; four red CatWISE counterparts remain, and none is detected at 12 μm, where an enclosure's energy would appear, consistent with brown dwarfs and obscured quasars aligned by chance.

Breakthrough Listen targets. Three of the 170 stars exceed the colour criterion: HR 8799, whose debris disc is known, a blended binary, and a star with a catalogued 22 μm excess.

SPHEREx spectra. The SPHEREx spectra of the Project Hephaistos candidates show excesses only where a background galaxy (candidates D and E) or a radio source (candidate G) lies within the WISE beam. JWST imaging of candidates D and E has since resolved the galaxies, within about 1″ of each star, as the source of their mid-infrared excess [10], which SPHEREx's 6″ pixels could attribute only by the spectral shape. For the Contardo and Hogg stars, the ratio to the controls shows a 4 to 5 μm excess above 3σ, with a systematic floor of 0.063 from 50 control pairs, in 37 of 51 and a 3 to 4 μm excess in 36, against 1 and 0 control pairs. A black body puts every excess between 486 and 2133 K holding 0.6 to 7 per cent of the star's light: 22 at 400 to 1000 K, as warm dust, and nine above 1700 K, as cool companions or blends; none lies below 400 K, and no 3.3 μm aromatic band reaches 2σ. The absence of cooler components is a property of the spectra's reach as much as of the stars: a 300 K body emits at 5 μm a hundredth of what it emits at its 10 μm peak, so the fit from 1.6 to 5 μm is sensitive to components above about 400 K, and a cooler component that the 12 and 22 μm WISE excesses of these stars might hold is not excluded. In the 41 stars that the Legacy Surveys cover, the excess sits in the star's own deblended source, whose $W2/W1$ ratio exceeds its controls' by up to 0.85, and no source within 8″ carries more than 6 per cent of it; the unWISE $W2$ centroid of every star lies within 3σ of its $W1$ centroid against the controls' 0.12″ scatter, which leaves only a red source within about 0.5″. The eleven with the largest excesses keep them at every NEOWISE visit from 2014 to 2024, with a visit-to-visit scatter of 0.02 to 0.06 mag in $W1 - W2$.

4. Related work

Waste-heat searches. Carrigan's IRAS search set whole-sky limits on spheres at 100 to 600 K [2], and the $\hat{G}$ search found no galaxy re-radiating most of its starlight [3, 4]. Project Hephaistos selected partial-sphere candidates among Milky Way stars with Gaia, 2MASS and WISE [5, 6], and radio and optical images later showed background sources near several of them [7], as the SPHEREx spectra here do. Three further studies of those candidates have appeared: archival diagnostics find background contamination in candidates B and C and suggestive signs in five more [8], their stellar parameters are those of ordinary M dwarfs whose infrared colours resemble transitional discs without the signs of youth [9], and JWST resolves background galaxies within about 1″ of candidates D and E [10], the two whose SPHEREx spectra attribute the excess to a galaxy here. Contardo and Hogg modelled the mid-infrared colours of five million FGK stars from their Gaia spectra and published the most extreme excesses [11].

Companions and vanishing sources. Kervella et al. derived companion masses from the Hipparcos-Gaia proper-motion anomaly [14], and the VASCO project catalogued Palomar sources missing from modern surveys [21, 20]; we find no earlier use of either as a test for enclosed stars.

5. Conclusion

No nearby star shows the infrared emission of a Dyson sphere in these tests. Optically dark companions of known mass re-radiate at most a few tenths of a per cent of a star's light at 300 K, habitable-zone hosts carry no mean warm excess at the level of 10^{-5} of their light, no star within 15 pc brightened over a decade, and no star that vanished from the Palomar sky reappears in the mid-infrared. The candidate excesses that SPHEREx resolves are background sources, warm dust and cool companions. The last of the companions' excesses, 3.3σ at 12 μm in HD 59099, is absent from the WISE All-Sky release of the same exposures.

The limits apply to structures re-radiating as black bodies at the temperatures tested, and to the epochs and sensitivities of the surveys: a 300 K enclosure of a star of 0.01 $L_\odot$ within 210 pc moving faster than 19 to 40 km s^{-1} across the line of sight, a 30 to 100 K shell holding more than 0.2 per cent of a solar luminosity at 50 pc, and an enclosure at 200 to 600 K of a vanished source brighter than $R \approx 19$. A sphere at a few tens of kelvin around a star beyond 50 pc, one that emits mostly outside the WISE and AKARI bands, a slow-moving enclosed star, or one whose waste heat is directed away from the Sun is not constrained, and the SPHEREx fits do not reach components below 400 K. SPHEREx's spectra of the dark companions and of the habitable-zone hosts would extend the stacked test to 5 μm with resolved spectral shapes.

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